

Electronics: Mini Project 1 Report

Kuhu Jayaswal

September 16, 2026

1 Introduction

This miniproject involved designing a compact, USB-powered LED flasher with a period within $\pm 10\%$ of 1 s. USB VBUS is regulated from 5 V to 3.3 V, and an MCP6022-based Schmitt-trigger relaxation oscillator generates the flashing signal. The circuit and PCB were designed in KiCad, while LTspice and MATLAB were used to evaluate the oscillator period under component tolerances.

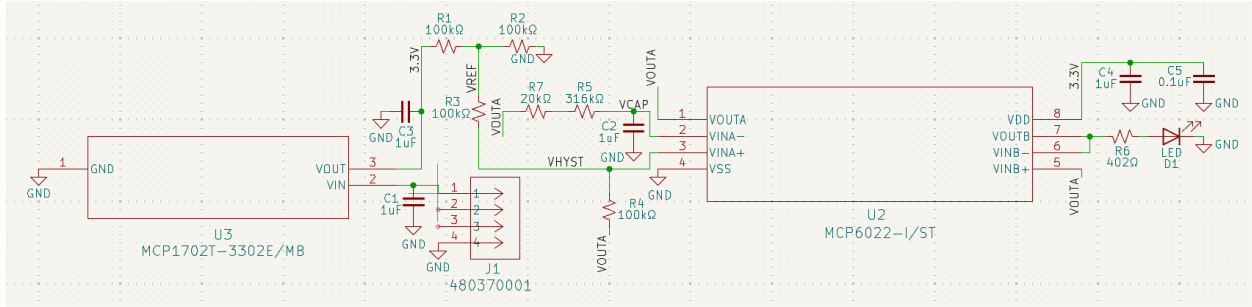


Figure 1: Final KiCad schematic of the USB-powered LED flasher.

2 Design Choices

2.1 Power Supply

The 5 V USB VBUS supply is regulated to 3.3 V using an MCP1702-3302 LDO. The regulator uses 1 μF input and output capacitors for stability and filtering. The MCP6022 is additionally bypassed with 1 μF and 0.1 μF capacitors, with the 0.1 μF capacitor placed close to its supply pins.

2.2 Schmitt-Trigger Relaxation Oscillator

One amplifier of the MCP6022 is configured as a Schmitt-trigger relaxation oscillator. Two 100 k Ω resistors create a midpoint reference voltage:

$$V_{\text{REF}} = 3.3 \left(\frac{100 \text{ k}\Omega}{100 \text{ k}\Omega + 100 \text{ k}\Omega} \right) = 1.65 \text{ V}. \quad (1)$$

Positive feedback creates hysteresis around V_{REF} , producing upper and lower switching thresholds. Meanwhile, the timing capacitor at the inverting input charges and discharges through the timing resistance. Each time V_{CAP} crosses a switching threshold, the op-amp changes state, creating a continuous square-wave output.

2.3 Timing Network

A 1 μF capacitor was selected as the timing capacitor. The final timing resistance uses two available resistor values in series:

$$R_T = 316 \text{ k}\Omega + 20 \text{ k}\Omega = 336 \text{ k}\Omega. \quad (2)$$

This combination placed the nominal oscillation period near 1 s while providing margin from the 0.9 s and 1.1 s specification limits. Component variation was then evaluated using Monte Carlo simulation.

2.4 Output Buffer and LED

The second MCP6022 amplifier is configured as a unity-gain buffer so that the LED does not directly load the oscillator. A 402 Ω series resistor limits the LED current. Using the red LED's 1.5–2.4 V forward voltage range gives

$$I_{\text{LED}} \approx 2.24\text{--}4.48 \text{ mA}. \quad (3)$$

This provides a visible output while keeping the op-amp output current relatively low.

2.5 PCB Layout

The circuit was implemented as a two-layer PCB with all components on the top side. Ground pours were used on both copper layers and connected with ground vias. The timing and hysteresis components were placed close to the MCP6022 to keep sensitive analog connections short, while bypass capacitors were placed close to the IC supply pins. The board was kept compact while meeting the required trace-width, spacing, and via constraints.

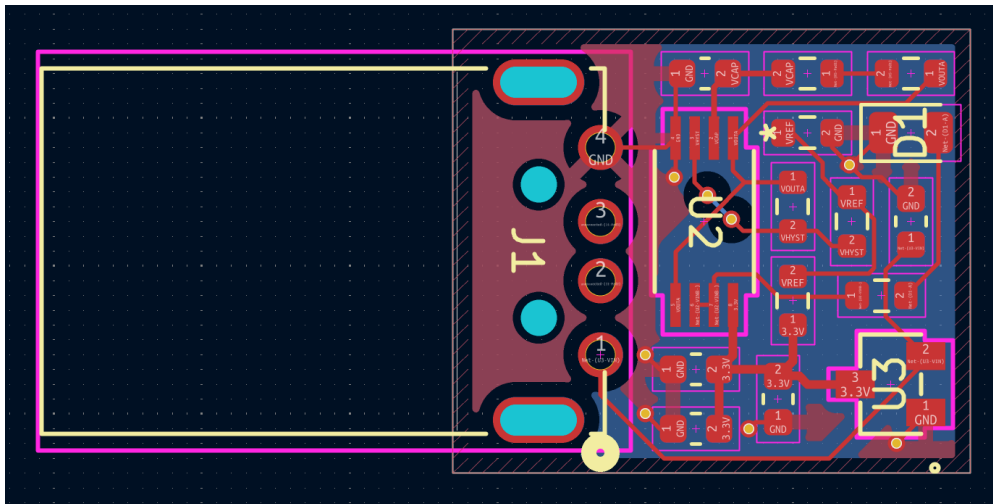


Figure 2: Final two-layer PCB layout in KiCad.

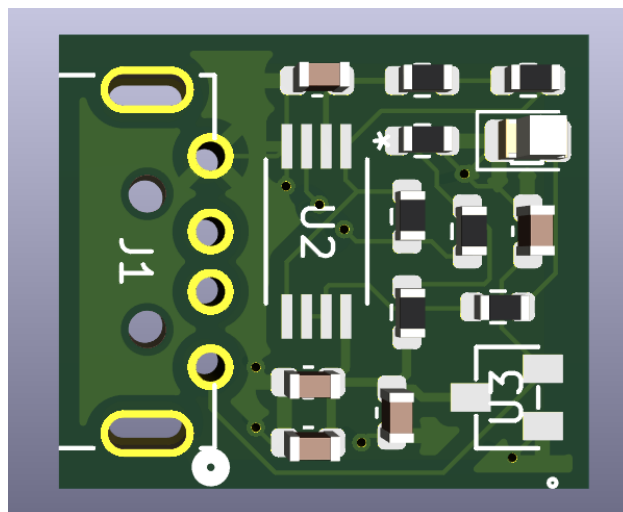


Figure 3: KiCad 3-D view of the completed PCB layout.

3 Monte Carlo Tolerance Analysis

The oscillator period depends on the resistor and capacitor values, so component tolerances produce variation in the flashing period. A 250-run Monte Carlo transient simulation was performed in LTspice to evaluate this variation. The 316 k Ω and 20 k Ω timing resistors were modeled as separate physical components, and MATLAB was used to extract the period from successive rising edges of each simulated output waveform.

The resulting period distribution was

$$T_{\text{mean}} = 0.9856 \text{ s}, \quad T_{\text{min}} = 0.9339 \text{ s}, T_{\text{max}} = 1.0314 \text{ s}, \sigma_T = 0.0183 \text{ s}. \quad (4)$$

All 250 simulated trials satisfied the required period specification:

$$0.90 \text{ s} \leq T \leq 1.10 \text{ s}. \quad (5)$$

The simulation therefore supports that the selected component values meet the period requirement under the modeled component variations.

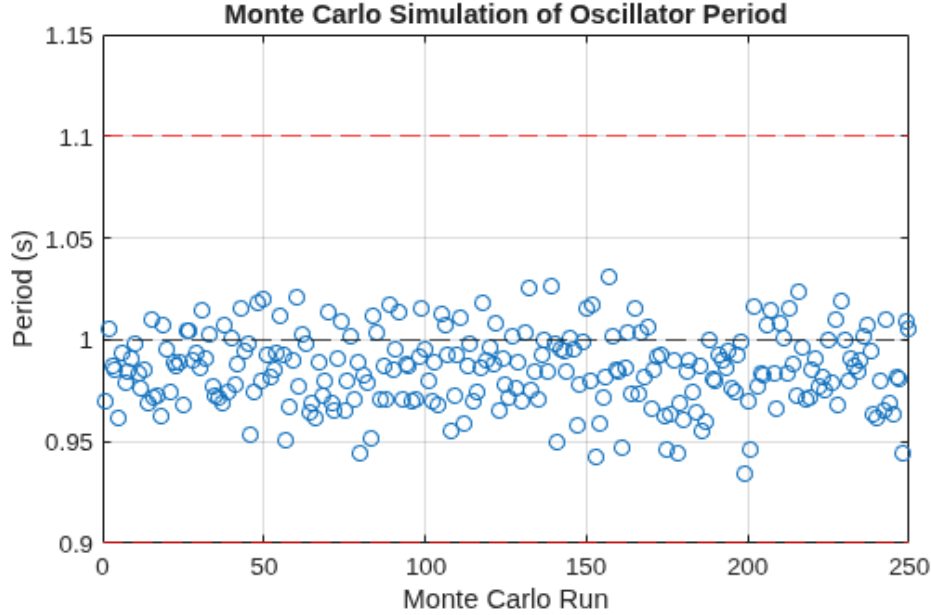


Figure 4: Oscillation period measured for 250 Monte Carlo trials. The dashed red lines indicate the 0.9 s and 1.1 s and the dashed black line indicates the 1 s target period.

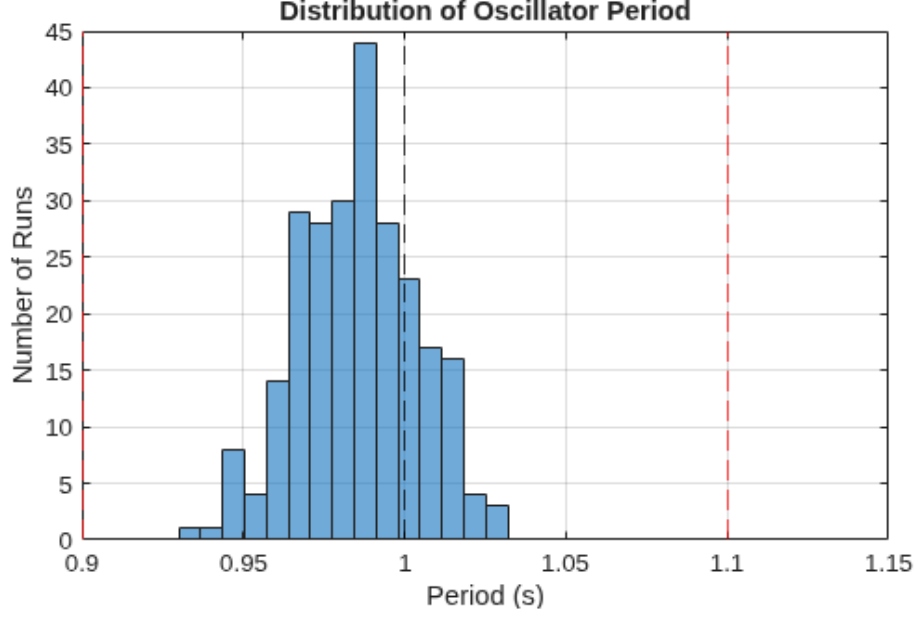


Figure 5: Distribution of the oscillator period across the 250 Monte Carlo trials.

4 Bill of Materials and Design Files

4.1 BOM

Table 1 lists the components used in the final design.

Ref.	Qty.	Value	Description	DigiKey Catalog #	Manufacturer Part #
C1-C4	4	1 μ F	CAP CER 1UF 25V X7R 0603	399-7376-1-ND	C0603C105K3RACTU
C5	1	0.1 μ F	CAP CER 0.1UF 50V X7R 0603	311-1779-1-ND	CC0603JRX7R9BB104
R1-R4	4	100 k Ω	RES SMD 100K OHM 1% 1/10W 0603	311-100KHRCT-ND	RC0603FR-07100KL
R5	1	316 k Ω	RES SMD 316K OHM 1% 1/10W 0603	311-316KHRCT-ND	RC0603FR-07316KL
R6	1	402 Ω	RES SMD 402 OHM 1% 1/10W 0603	311-402HRCT-ND	RC0603FR-07402RL
R7	1	20 k Ω	RES SMD 20K OHM 1% 1/10W 0603	311-20.0KHRCT-ND	RC0603FR-0720KL
D1	1	Red LED	LED RED CLEAR 0805 SMD	160-1427-1-ND	LTST-C171KRKT
U2	1	MCP6022	IC OPAMP GP 10MHZ RRO 8TSSOP	MCP6022T-I/STCT-ND	MCP6022T-I/ST
U3	1	3.3 V LDO	IC REG LIN 3.3V 250MA SOT23A-3	MCP1702T-3302E/CBCT-ND	MCP1702T-3302E/CB
J1	1	USB Type-A	CONN PLUG USB 4POS RT ANG PCB	WM17117-ND	0480370001

Table 1: BOM for the USB-powered LED flasher taken from the provided parts list.

4.2 Design Files

The repository includes the KiCad schematic and PCB layout, LTspice simulation files, MATLAB analysis code, and additional project files used for the design.

<https://github.com/k-jayaswal/Electronics-olin/tree/main/mp1>

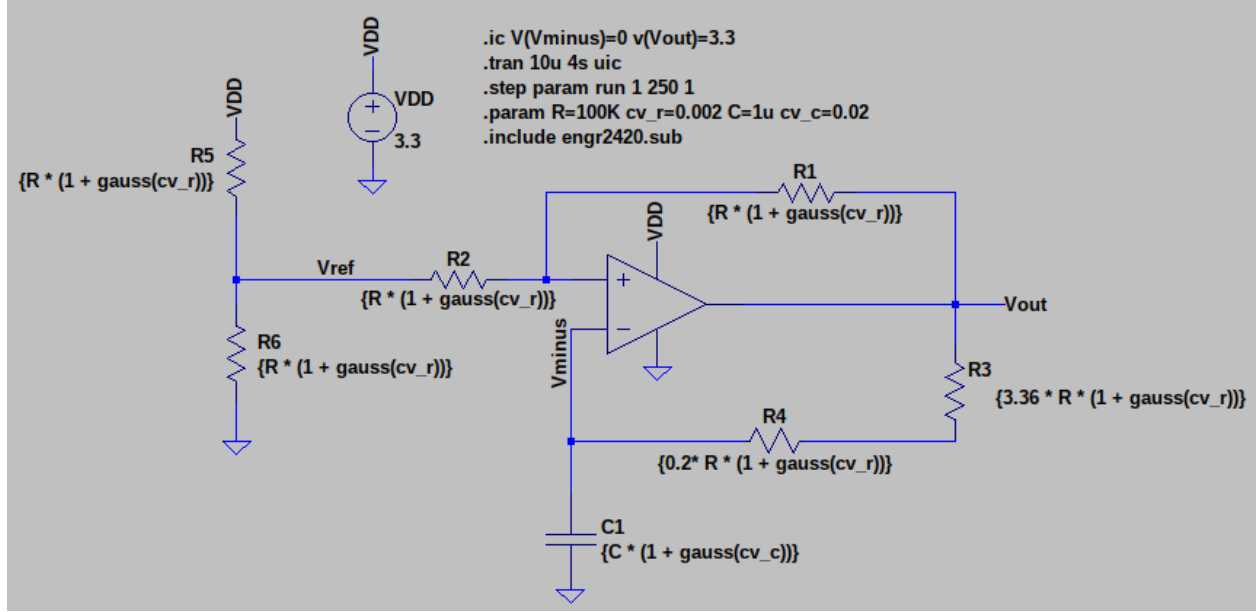


Figure 6: LTspice model of the Schmitt-trigger relaxation oscillator used for the Monte Carlo tolerance analysis.

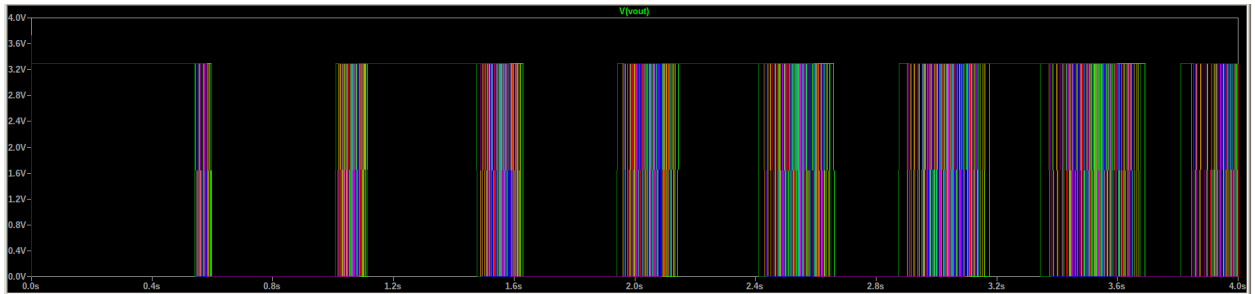


Figure 7: LTspice transient output showing the periodic oscillator output voltage, V_{out} .